

RESEARCH METHODOLOGY – NOTEBOOK MATERIAL

Project Title

AI-Enabled Framework for Detection of Red Palm Weevil Infestation in Coconut Trees

Problem Statement

To identify Red Palm Weevil (RPW) infestation and assess the severity under real-world field conditions using Artificial Intelligence (AI) that can identify RPW infestation at an early stage and assess its severity, thereby supporting timely and informed pest-management decisions.

Literature-review sub-area

Importance and Impact of Artificial Intelligence in Detection of Red Palm Weevil in Coconut

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2. PROJECT TITLE AND PROBLEM STATEMENT

Project Title: AI-Enabled Framework for Detection of Red Palm Weevil Infestation in Coconut Trees.

Problem Statement: To identify Red Palm Weevil (RPW) infestation and assess the severity under real-world field conditions using Artificial Intelligence (AI) that can identify RPW infestation at an early stage and assess its severity, thereby supporting timely and informed pest-management decisions.

Objective of the study: Develop a literature-grounded AI framework that combines relevant sensing and AI techniques for early RPW detection, severity assessment, and practical field deployment.

3. IMPORTANCE AND IMPACT OF ARTIFICIAL INTELLIGENCE

- **Early detection:** AI can learn patterns from acoustic signals and visual images that may indicate infestation before severe visible damage becomes obvious.
- **Automated analysis:** Machine-learning/deep-learning models can process large numbers of observations consistently and reduce dependence on manual inspection.
- **Multimodal evidence:** Audio can capture larval activity while RGB images can capture adult insects and visible symptoms; combining modalities can provide complementary evidence.
- **Severity assessment:** AI can classify or score infestation status and severity classes when suitable labelled data are available.
- **Field decision support:** A lightweight model can support timely inspection and pest-management decisions at tree or farm level.
- **Scalability:** IoT/edge devices and mobile/field sensing can enable repeated monitoring rather than one-time inspection.

Impact: The literature indicates that AI is moving RPW detection from manual/subjective inspection toward automated, sensor-assisted and data-driven monitoring. The proposed framework extends this direction by considering bimodal audio + RGB evidence and farm-level spatial-temporal monitoring.

4. LITERATURE REVIEW – 18 PAPERS

No.	Paper / Year	Main method	Relevance to proposed work
1	Soroker et al., 2010	Trunk-embedded acoustic sensors	Captured larval sound activity inside palms; establishes acoustic sensing as a basis for RPW

No.	Paper / Year	Main method	Relevance to proposed work
2	Faleiro et al., 2012	Field acoustic validation	Studied acoustic detection under field conditions and environmental noise; important for robustness.
3	Martin, Saranya & Chowdary, 2026	IoT + Raspberry-Pi thermal CNN	Thermal imaging and CNN pipeline for RPW-related detection; demonstrates edge/IoT directio
4	Alsanea et al., 2022	R-CNN visual localisation	Used R-CNN-based computer vision for locating RPW/related visual evidence.
5	Kagan et al., 2021	Aerial/street imagery	Landscape-scale palm mapping; relevant to large-area monitoring.
6	Rigakis et al., 2021	TreeVibes acoustic dataset	Public labelled acoustic RPW dataset/platform; useful for audio model development.
7	Kurdi et al., 2021	Classical ML benchmark	Curated RPW dataset evaluated using conventional machine-learning approaches.
8	Boulila et al., 2023	Sound-to-spectrogram CNN	Converted acoustic signals into spectrogram images for CNN classification.
9	Wu et al., 2023	Attention-enhanced YOLOv5	Approximately 2,600 RGB images; attention-enhanced YOLOv5 for visual detection.
10	Nadeem et al., 2025	Paired RGB + thermal	832 RGB and thermal date-palm images with farmer-informed health classes.
11	Deng et al., 2025	AMS-YOLO	YOLOv8n with SMCA attention, MSBlock and AMConv for improved detection.
12	Yang et al., 2023	Maize-YOLO	YOLOv7-based lightweight/modified detector; useful as an agricultural object-detection design
13	Lu et al., 2023	CNN-BiGRU	CNN + BiGRU architecture for agricultural disease identification; supports temporal feature lea
14	Kamalesh Kanna et al., 2024	YOLO comparative review	Compared YOLO versions in agriculture; useful for selecting a detector family.
15	Wang et al., 2021	Fibre-optic DAS + ANN/CNN	Acoustic/vibration sensing with ANN/CNN; reported >99% accuracy in its study setting.
16	Susan T et al., 2026 (RED-BIYO)	YOLOv9 + SCNN-BiGRU + fuzzy fusion	Bimodal stem-image and audio pipeline; Mamdani fuzzy fusion; reported 99.54% accuracy.
17	Stephen, Martin & Rajesh, 2026	Acoustic + microwave + ASKF	Attention-guided acoustic classifier + sensor-guided transformer; adaptive Kalman fusion; rep
18	Saranya & Martin, 2025	LPC + cosine similarity + Bi-LSTM	Acoustic classifier using LPC and Bi-LSTM; reported 98.02% accuracy.

Literature observation: The papers collectively show three major directions: acoustic sensing, visual/thermal computer vision, and multimodal sensor fusion. The proposed framework is positioned at the intersection of acoustic + RGB detection with lightweight field deployment and spatial-temporal monitoring.

5. DATASET DETAILS

Dataset 1 – TreeVibes: A labelled trunk-mounted acoustic dataset containing recordings from infested/non-infested palms and other borer-susceptible trees across species, locations and seasons. It is audio-focused and is not a same-tree paired RGB dataset.

Dataset 2 – Red Palm Weevil Object Detection Dataset: A community-maintained RGB image dataset with bounding boxes for weevils and/or infestation symptoms and YOLO-compatible exports.

Dataset 3 – Nadeem et al. (2025): 832 RGB and thermal date-palm images. Labels are based on farmers’ feedback and include non-infected, infected, badly damaged and dead classes.

Important dataset gap: TreeVibes audio and the public RGB object-detection data are not inherently paired observations from the same tree. Therefore, direct multimodal fusion requires either a purpose-built paired subset or a carefully designed late-fusion strategy.

Dataset details to record during implementation: number of samples, class distribution, acquisition location, season, sensor/device, sampling rate for audio, image resolution, annotation format, train/validation/test split and whether the split is tree-level rather than clip-level.

6. PERFORMANCE METRICS AND MATHEMATICAL FORMULAS

Confusion-matrix terms: TP = true positive, TN = true negative, FP = false positive, FN = false negative.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall / Sensitivity / TPR} = TP / (TP + FN)$$

$$\text{Specificity} = TN / (TN + FP)$$

$$\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$\text{MCC} = (TP \times TN - FP \times FN) / \sqrt{[(TP + FP)(TP + FN)(TN + FP)(TN + FN)]}$$

ROC-AUC: Area under the True Positive Rate versus False Positive Rate curve across decision thresholds.

mAP: Mean Average Precision across the evaluated object-detection classes.

IoU: Intersection over Union = area of overlap between predicted and ground-truth boxes / area of their union.

Why multiple metrics? Accuracy alone can be misleading when field data are imbalanced. Precision, recall, specificity, F1 and MCC provide complementary information. Object detection additionally requires IoU and mAP.

7. EXPERIMENTAL SETUP AND EXPERIMENT DETAILS

RED-BIYO (Paper 16): 3,000 stem images and 2,400 acoustic clips. Reported 80/10/10 partition per fold, 5-fold cross-validation, class balancing and fixed seed. Hardware: Intel i7, 16 GB RAM and NVIDIA RTX 3060. Reported 99.54% accuracy, MCC 97.93% and F1 94.87%.

Acoustic–Microwave Fusion (Paper 17): 3,077 acoustic clips (1,332 RPW / 1,745 non-RPW) and 2,000 microwave samples (1,000 / 1,000). Used an 80/10/10 stratified split without k-fold cross-validation and compared four fusion strategies. Acoustic-only was reported at 99%, microwave-only at 50%, and adaptive sequential Kalman-filter fusion at 96.2%, with AUC 0.98.

LPC + Bi-LSTM (Paper 18): 3,000 acoustic segments (1,500 RPW / 1,500 non-RPW); 2,400 training and 600 test samples. Single held-out split, 3,050 training iterations and validation every 10 iterations. MATLAB implementation with band-pass filtering, LPC order 14 and Bi-LSTM. Reported 98.02% accuracy, precision 0.972, recall 96.65% and F1 97%.

Wu et al. (Paper 9): Approximately 2,600 RGB images with a 7:2:1 train/validation/test split.

Proposed experiment plan: Use tree-level separation to prevent leakage between clips/images from the same tree. Compare audio-only, RGB-only and fused models. Report accuracy, precision, recall, specificity, F1, MCC and ROC-AUC; for detection also report IoU/mAP. Record model size and inference latency on an edge-class device.

8. RESEARCH GAP

- **Modality pairing gap:** Public audio and RGB datasets are generally not same-tree paired, limiting direct multimodal training.
- **Single-tree focus:** Many studies make an individual-tree decision; fewer explicitly model spread across neighbouring trees.
- **Deployment gap:** Some studies use workstation GPUs, while field use requires lightweight models and measured edge inference latency.
- **Evaluation gap:** Accuracy is often emphasized; field-oriented evaluation should also report specificity, MCC and class-wise performance.
- **Fusion gap:** Fuzzy rules are interpretable but can be manually defined; confidence-weighted/adaptive fusion can use modality reliability more directly.

Proposed research direction: A lightweight bimodal audio + RGB framework with confidence-weighted fusion, followed by spatial-temporal analysis of tree-level confidence to identify emerging clusters.

9. PROPOSED AI FRAMEWORK / ALGORITHM

Algorithm name: Bimodal Audio–Visual AI Framework for Early RPW Detection, Severity Assessment and Farm-Level Monitoring.

Step 1 – Audio acquisition: Capture trunk acoustic signals from the tree.

Step 2 – Audio preprocessing: Band-pass filtering, framing and windowing; convert the signal into a spectrogram using STFT.

Step 3 – Audio AI: CNN extracts local spectral features; BiGRU models temporal dependencies; attention emphasizes informative regions.

Step 4 – RGB acquisition: Capture an RGB image of the palm/stem.

Step 5 – Visual AI: Use a lightweight YOLO-family detector to identify adult weevil or visible symptoms such as boreholes, frass and discoloration.

Step 6 – Confidence fusion: Combine audio and visual probabilities using confidence-weighted fusion. A Mamdani fuzzy system is an interpretable alternative.

Step 7 – Severity: Map fused confidence/features to healthy, infected/early and higher-severity categories according to the final labelled dataset.

Step 8 – Spatial graph: Represent trees as graph nodes and connect nearby trees.

Step 9 – Spatial propagation: Smooth confidence using neighbouring-tree evidence.

Step 10 – Temporal smoothing: Combine current and previous observations to reduce unstable alerts.

Step 11 – Farm-level alert: Detect connected high-confidence clusters and issue an early outbreak alert.

10. MATHEMATICAL STEPS

Hamming window: $w(n) = 0.54 - 0.46 \cos(2\pi n/(N-1))$; $xw(n) = x(n)w(n)$.

STFT: $\text{STFT}(m,k) = \sum xw(n)e^{-j2\pi kn/N}$; spectrogram magnitude $S(m,k) = |\text{STFT}(m,k)|$.

Min-max normalization: $S_{\text{norm}} = (S - S_{\text{min}})/(S_{\text{max}} - S_{\text{min}})$.

Attention: $\text{Attention}(Q,K,V) = \text{Softmax}(QK^T/\sqrt{d_k})V$.

Confidence fusion: $C_i = w_{\text{audio}}P_{\text{audio}} + w_{\text{visual}}P_{\text{visual}}$, where $w_{\text{audio}} + w_{\text{visual}} = 1$.

Example: $P_{\text{audio}}=0.80$, $P_{\text{visual}}=0.90$, $w_{\text{audio}}=0.40$, $w_{\text{visual}}=0.60 \rightarrow C=0.40(0.80)+0.60(0.90)=0.86$.

Spatial graph: $G=(V,E)$; for nearby trees $e_{ij} = \exp(-d(i,j)^2/(2\sigma^2))$, otherwise $e_{ij}=0$.

Spatial propagation: $C_i(t+1)=\alpha C_i(t)+(1-\alpha)[\sum_j e_{ij}C_j(t)/\sum_j e_{ij}]$.

Temporal smoothing: $C_i(t)=\beta C_i(t)+(1-\beta)C_i(t-1)$.

Outbreak score: $O(t)=\sum_j 1[C_i(t)>\theta_{\text{low}}]$, with connected clusters considered and a minimum cluster size k .

Alerts: tree-level alert if $C_i(t)>\theta_{\text{high}}$; farm-level alert if $O(t)>\theta_{\text{cluster}}$.

11. EXAMPLE PAPER STUDY

Example: RED-BIYO (Susan T et al., 2026)

Problem addressed: RPW detection using complementary image and acoustic information.

Inputs: stem images + acoustic clips.

Models: YOLOv9 for image processing and SCNN-BiGRU for acoustic processing.

Fusion: Mamdani fuzzy logic; the report notes a crisp threshold where output >0.6 triggers “Not-Healthy”.

Experimental design: 5-fold cross-validation with class balancing and fixed seed; 80/10/10 split per fold.

Reported performance: 99.54% accuracy, MCC 97.93%, F1 94.87%.

Research relevance: This paper directly supports the proposed bimodal design. The proposed work further emphasizes lightweight deployment and spatial-temporal farm-level outbreak monitoring.

Important note: Reported metrics belong to the cited paper's dataset and experimental conditions; they should not be presented as expected or guaranteed performance of the proposed project.

12. GRAPHS / FLOW AND THEIR EXPLANATION

Graph 1 – Confusion matrix: A 2×2 table showing TP, TN, FP and FN. It is used to calculate classification metrics.

Graph 2 – ROC curve: Plot of TPR against FPR for different thresholds. AUC summarizes discrimination across thresholds.

Graph 3 – Precision–Recall curve: Useful when positive cases are relatively uncommon or class imbalance matters.

Graph 4 – Training/validation curves: Accuracy and loss versus epochs/iterations. A widening train–validation gap may indicate overfitting.

Graph 5 – Detection example: RGB image with predicted bounding boxes and confidence scores around RPW or symptoms.

Graph 6 – Farm spatial graph: Trees are nodes; edges represent nearby trees. High-confidence connected nodes can reveal an emerging infestation cluster.

Flow to draw in notebook: Tree → Audio Sensor → Preprocessing → Spectrogram → CNN + BiGRU + Attention → P_audio; AND Tree → RGB Camera → YOLO Detector → P_visual → Confidence Fusion → Severity → Spatial Graph → Temporal Smoothing → Tree/Farm Alert.

13. AI RELEVANCE AND EXPECTED IMPACT

- AI automates feature extraction and pattern recognition from high-dimensional acoustic and image data.
- Deep learning reduces the need to manually define every visual or acoustic feature.
- Multimodal AI can use complementary evidence and reduce dependence on one sensing mode.
- Spatial-temporal AI can shift the objective from isolated detection to early identification of emerging infestation clusters.
- Edge deployment can support repeated field monitoring with lower dependence on cloud connectivity.

Expected outcome: An AI-assisted decision-support framework for earlier RPW identification, severity assessment and targeted intervention.

14. SDG RELEVANCE

SDG 9 – Industry, Innovation and Infrastructure: The project applies AI, sensing, computer vision, deep learning and potential IoT/edge computing to build digital agricultural monitoring infrastructure. The lightweight deployment objective is particularly relevant to practical technological infrastructure.

SDG 15 – Life on Land: RPW can damage palm ecosystems and agricultural resources. Earlier detection can support timely control and reduce avoidable tree loss and spread, aligning the project with technology-assisted protection of terrestrial ecosystems.

15. CONCLUSION

The literature demonstrates that RPW detection has progressed from acoustic sensing and conventional machine learning to deep-learning-based visual detection and multimodal fusion. The updated project framework, “AI-Enabled Framework for Detection of Red Palm Weevil Infestation in Coconut Trees,” focuses on early detection and severity assessment under real-world conditions. Its proposed contribution is to combine acoustic and RGB evidence, use confidence-aware fusion, and extend tree-level predictions into spatial-temporal farm-level monitoring while considering edge deployment and robust evaluation metrics.

16. NOTEBOOK CHECKLIST

- Index on first page: Section No. / Topic / Page No.
- Write exact updated project title and problem statement.
- Write importance and impact of AI.
- Include literature review and explain the selected example paper.
- Write dataset source/details, class information and split strategy.
- Write performance metrics with formulas and explain TP/TN/FP/FN.
- Write experimental setup: data size, split, hardware/software, training and evaluation.
- Clearly write research gap.
- Clearly identify the proposed algorithm and explain each step.
- Write mathematical formulas used in the algorithm.

- Draw and explain graphs/flow diagrams.
- Mention AI relevance and SDG 9/SDG 15.
- Do not copy reported paper accuracy as the performance of your own proposed system.

17. REFERENCES USED IN THE STUDY

- [1] Soroker et al., 2010 – acoustic detection of RPW larvae.
- [2] Faleiro et al., 2012 – field acoustic validation.
- [3] Martin, Saranya & Chowdary, 2026 – IoT/Raspberry-Pi thermal CNN pipeline.
- [4] Alsanea et al., 2022 – R-CNN visual RPW localisation.
- [5] Kagan et al., 2021 – landscape-scale palm mapping.
- [6] Rigakis et al., 2021 – TreeVibes acoustic dataset/platform.
- [7] Kurdi et al., 2021 – curated RPW ML dataset benchmark.
- [8] Boulila et al., 2023 – spectrogram-based CNN classification.
- [9] Wu et al., 2023 – attention-enhanced YOLOv5 RGB detection.
- [10] Nadeem et al., 2025 – paired RGB/thermal date-palm dataset.
- [11] Deng et al., 2025 – AMS-YOLO.
- [12] Yang et al., 2023 – Maize-YOLO.
- [13] Lu et al., 2023 – CNN-BiGRU agricultural disease identification.
- [14] Kamalesh Kanna et al., 2024 – YOLO comparison in agriculture.
- [15] Wang et al., 2021 – fibre-optic DAS + ANN/CNN RPW classification.
- [16] Susan T et al., 2026 – RED-BIYO multimodal RPW detection.
- [17] Stephen, Martin & Rajesh, 2026 – acoustic/microwave adaptive Kalman fusion.
- [18] Saranya & Martin, 2025 – LPC + Bi-LSTM acoustic classifier.

Dataset sources recorded in the study: TreeVibes (Kaggle) and Red Palm Weevil Object Detection Dataset (Roboflow Universe).